

BaraGo: Integration of Automated Appointment and Healthcare Scheduling Management System

1. Project Overview

BaraGo is a web-based healthcare scheduling management system designed for barangay use. It allows residents to book checkup appointments, request ambulance assistance, and receive notifications once their requests are approved, rejected, or rescheduled.

The system helps reduce wasted visits to the barangay health center by allowing residents to know if checkup schedules are available before going there. It also helps barangay health staff manage appointment requests, ambulance requests, resident records, schedules, and printable reports in a more organized way.

2. Problem Statement

In some barangays, residents go to the health center without knowing if checkups are available. This can waste their time and effort, especially when no health worker or checkup service is available on that day.

There are also residents who may need emergency assistance or ambulance support but cannot easily go to the barangay office to request help. Because of this, there is a need for a system that can help residents request health services online and help barangay staff manage those requests properly.

3. Project Objective

The main objective of **BaraGo** is to provide a useful and organized system for barangay healthcare services.

Specifically, the system aims to:

1. Allow residents to book checkup appointments online.

2. Allow residents to request ambulance assistance.
3. Help barangay staff approve, reject, or reschedule requests.
4. Notify residents about their appointment or ambulance request status.
5. Help barangay staff manage health schedules.
6. Provide searchable records and printable reports.
7. Improve public service management in the barangay.

4. Target Users

Resident

Residents are the main users of the system. They can register, log in, book appointments, request ambulance assistance, view notifications, and check their appointment history.

Barangay Health Staff / Admin

The admin manages the system. They can approve or reject requests, create health schedules, manage resident records, and generate reports.

Health Worker

Health workers can view approved appointments, update checkup status, and add remarks after the consultation.

5. MVP

The **Minimum Viable Product** of BaraGo should include the most important features needed for the system to work.

Resident MVP Features

1. Register and log in
2. View resident dashboard
3. Book a checkup appointment
4. Request ambulance assistance
5. View request status

6. Receive notifications
7. View appointment history

Admin MVP Features

1. Admin login
2. View dashboard summary
3. Manage appointment requests
4. Manage ambulance requests
5. Manage health schedules
6. Manage resident records
7. Generate printable reports

6. Main Features

Resident Features

1. Registration and Login

Residents can create an account using their personal information.

Required details:

Field	Description
Full Name	Resident's complete name
Birthdate	Date of birth
Age	Resident's age
Gender	Male/Female/Other
Address	Complete barangay address
Purok	Resident's purok
Contact Number	Used for updates

Email / Username	Login credential
Password	Account security

2. Resident Dashboard

After logging in, the resident can see:

Section	Purpose
Upcoming Appointment	Shows the next approved checkup
Appointment Status	Pending, approved, rejected, or completed
Ambulance Request Status	Shows emergency request progress
Notifications	Shows updates from barangay staff
Quick Actions	Book appointment or request ambulance

3. Book Checkup Appointment

Residents can request a schedule for checkup.

Form fields:

Field	Description
Patient Name	Name of the patient
Patient Age	Age of the patient
Reason for Checkup	Fever, follow-up, consultation, etc.
Preferred Date	Desired checkup date
Preferred Time	Morning or afternoon
Notes	Additional concern

Appointment status flow:

Pending → Approved / Rejected / Rescheduled → Completed

4. Ambulance Request

Residents can request ambulance assistance if the patient cannot walk or needs urgent transport.

Form fields:

Field	Description
Patient Name	Name of the patient
Exact Location	Location where ambulance should go
Contact Number	Emergency contact
Emergency Type	Accident, cannot walk, severe pain, etc.
Description	Details of the emergency
Urgency Level	Low, medium, or high

Ambulance status flow:

Pending → Approved / Rejected / Dispatched → Completed

Important note:

For life-threatening emergencies, residents should still contact the emergency hotline immediately. BaraGo is for barangay ambulance request coordination.

5. Notifications

Residents will receive updates such as:

Notification Type	Example
Appointment Approved	Your checkup is scheduled on May 20 at 9:00 AM.

Appointment Rejected	Your request was rejected due to unavailable schedule.
Appointment Rescheduled	Your checkup was moved to May 21.
Ambulance Approved	Your ambulance request has been approved.
Ambulance Dispatched	The ambulance has been dispatched to your location.
Follow-up Reminder	Your next checkup is scheduled tomorrow.

6. Appointment History

Residents can view their past and current appointments.

Column	Description
Date	Appointment date
Type	Checkup or follow-up
Reason	Reason for visit
Status	Pending, approved, completed, cancelled
Remarks	Notes from admin or health worker

Admin Features

1. Admin Dashboard

The admin dashboard shows important system summaries.

Card	Description
Total Residents	Number of registered residents
Pending Appointments	Requests waiting for approval
Approved Appointments	Confirmed checkup schedules

Pending Ambulance Requests	Emergency requests waiting for action
Completed Checkups	Finished appointments

2. Appointment Management

Admin can:

1. View appointment requests
2. Approve appointment requests
3. Reject appointment requests
4. Reschedule appointments
5. Add remarks
6. Mark appointments as completed

Table columns:

Patient	Date	Reason	Status	Action
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3. Ambulance Request Management

Admin can:

1. View ambulance requests
2. Approve requests
3. Reject requests
4. Mark ambulance as dispatched
5. Mark request as completed
6. Add emergency response notes

Table columns:

Patient	Location	Emergency Type	Urgency	Status	Action
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4. Health Schedule Management

Admin can create available checkup schedules.

Fields:

Field	Description
Date	Available checkup date
Start Time	Start of checkup service
End Time	End of checkup service
Slot Limit	Number of patients allowed
Assigned Staff	Doctor, nurse, or health worker
Status	Open, closed, or cancelled

5. Resident Management

Admin can:

1. View registered residents
2. Search residents
3. Verify resident accounts
4. Update resident details
5. Disable invalid accounts

6. Reports and Printable Records

Admin can generate and print:

Report	Purpose
Appointment List	Daily or weekly checkup list
Ambulance Request Report	Emergency request history
Resident List	Registered residents
Completed Checkup Report	Health center record
Monthly Summary Report	Barangay health service report

7. System Screens

Public Screens

1. Landing Page

Contains:

1. BaraGo logo
2. Short system description
3. Login button
4. Register button
5. Health service section
6. Ambulance assistance section

2. Login Page

Contains:

1. Email or username
2. Password
3. Login button
4. Register link

3. Register Page

Contains:

1. Full name
2. Birthdate
3. Age
4. Gender
5. Address
6. Purok
7. Contact number
8. Email or username
9. Password
10. Confirm password

Resident Screens

4. Resident Dashboard

Contains:

1. Welcome message
2. Upcoming appointment
3. Appointment status
4. Ambulance request status
5. Notifications
6. Quick action buttons

5. Book Appointment Page

Contains the checkup appointment request form.

6. Ambulance Request Page

Contains the ambulance assistance request form.

7. My Appointments Page

Shows all appointment requests and status.

8. Notifications Page

Shows updates from barangay staff.

9. Profile Page

Allows resident to view or update profile details.

Admin Screens

10. Admin Dashboard

Shows total residents, pending appointments, approved appointments, ambulance requests, and completed checkups.

11. Manage Appointments Page

Allows admin to approve, reject, reschedule, or complete appointment requests.

12. Manage Ambulance Requests Page

Allows admin to approve, reject, dispatch, or complete ambulance requests.

13. Manage Schedules Page

Allows admin to create, update, close, or cancel health schedules.

14. Manage Residents Page

Allows admin to search, verify, edit, or disable residents.

15. Reports Page

Allows admin to filter, generate, and print reports.

8. Branding

App Name

BaraGo

Meaning

BaraGo means **Barangay + Go**. It represents fast and accessible barangay services online.

Tagline

“Barangay health services, ready to go.”

Brand Personality

BaraGo should feel:

1. Simple
2. Trustworthy
3. Helpful
4. Community-centered
5. Clean and professional

Color Palette

Purpose	Color
Primary Green	#16A34A
Trust Blue	#2563EB
Background	#F8FAFC
Text	#111827
Emergency Red	#DC2626
Warning Yellow	#F59E0B

Logo Concept

The logo can combine:

1. A barangay house
2. A medical cross
3. A calendar check mark
4. A small arrow to represent “Go”

Logo idea:

A barangay house icon with a medical cross and a check mark, paired with the word BaraGo.

9. Database Schema

users

Stores login accounts.

```
CREATE TABLE users (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  full_name TEXT NOT NULL,  
  email TEXT UNIQUE NOT NULL,  
  password_hash TEXT NOT NULL,  
  role TEXT NOT NULL CHECK (role IN ('resident', 'admin', 'health_worker')),  
  status TEXT DEFAULT 'active' CHECK (status IN ('active', 'disabled')),  
  created_at TIMESTAMP DEFAULT NOW()  
);
```

residents

Stores resident profile details.

```
CREATE TABLE residents (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  user_id UUID REFERENCES users(id) ON DELETE CASCADE,  
  birthdate DATE,  
  age INT,  
  gender TEXT,  
  address TEXT NOT NULL,  
  purok TEXT,  
  contact_number TEXT NOT NULL,  
  emergency_contact_name TEXT,  
  emergency_contact_number TEXT,  
  verified BOOLEAN DEFAULT FALSE,  
  created_at TIMESTAMP DEFAULT NOW()  
);
```

health_schedules

Stores available checkup schedules.

```
CREATE TABLE health_schedules (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  schedule_date DATE NOT NULL,  
  start_time TIME NOT NULL,  
  end_time TIME NOT NULL,  
  slot_limit INT NOT NULL,  
  current_slots INT DEFAULT 0,  
  assigned_staff TEXT,
```

```
status TEXT DEFAULT 'open' CHECK (status IN ('open', 'closed', 'cancelled')),
created_at TIMESTAMP DEFAULT NOW()
);
```

appointments

Stores checkup appointment requests.

```
CREATE TABLE appointments (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  resident_id UUID REFERENCES residents(id) ON DELETE CASCADE,
  schedule_id UUID REFERENCES health_schedules(id) ON DELETE SET NULL,
  patient_name TEXT NOT NULL,
  patient_age INT,
  reason TEXT NOT NULL,
  preferred_date DATE,
  preferred_time TEXT,
  status TEXT DEFAULT 'pending' CHECK (
    status IN ('pending', 'approved', 'rejected', 'rescheduled', 'completed', 'cancelled')
  ),
  admin_remarks TEXT,
  created_at TIMESTAMP DEFAULT NOW(),
  updated_at TIMESTAMP DEFAULT NOW()
);
```

ambulance_requests

Stores ambulance assistance requests.

```
CREATE TABLE ambulance_requests (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  resident_id UUID REFERENCES residents(id) ON DELETE CASCADE,
  patient_name TEXT NOT NULL,
  exact_location TEXT NOT NULL,
  contact_number TEXT NOT NULL,
  emergency_type TEXT NOT NULL,
  description TEXT,
  urgency_level TEXT CHECK (urgency_level IN ('low', 'medium', 'high')),
  status TEXT DEFAULT 'pending' CHECK (
    status IN ('pending', 'approved', 'rejected', 'dispatched', 'completed', 'cancelled')
  ),
  admin_remarks TEXT,
  requested_at TIMESTAMP DEFAULT NOW(),
);
```

```
updated_at TIMESTAMP DEFAULT NOW()  
);
```

notifications

Stores updates sent to residents.

```
CREATE TABLE notifications (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  user_id UUID REFERENCES users(id) ON DELETE CASCADE,  
  title TEXT NOT NULL,  
  message TEXT NOT NULL,  
  type TEXT CHECK (type IN ('appointment', 'ambulance', 'system')),  
  is_read BOOLEAN DEFAULT FALSE,  
  created_at TIMESTAMP DEFAULT NOW()  
);
```

reports

Stores generated reports.

```
CREATE TABLE reports (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  generated_by UUID REFERENCES users(id) ON DELETE SET NULL,  
  report_type TEXT NOT NULL,  
  start_date DATE,  
  end_date DATE,  
  file_url TEXT,  
  created_at TIMESTAMP DEFAULT NOW()  
);
```

activity_logs

Tracks admin actions.

```
CREATE TABLE activity_logs (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  user_id UUID REFERENCES users(id) ON DELETE SET NULL,  
  action TEXT NOT NULL,  
  description TEXT,  
  created_at TIMESTAMP DEFAULT NOW()  
);
```

10. CRUD Coverage

Module	Create	Read	Update	Delete
Residents	Register account	View profile	Edit profile	Disable account
Appointments	Request appointment	View appointment	Reschedule/update status	Cancel request
Ambulance Requests	Submit request	View request	Update status	Cancel request
Health Schedules	Add schedule	View schedules	Edit schedule	Close/delete schedule
Notifications	Send notification	View notification	Mark as read	Delete notification
Reports	Generate report	View report	Filter report	Delete report

11. Final Short Description

BaraGo is a web-based barangay healthcare scheduling management system that helps residents book checkup appointments, request ambulance assistance, and receive notifications about their requests. It also helps barangay health staff manage schedules, approve or reject requests, monitor ambulance assistance, and generate printable records for organized public service management.